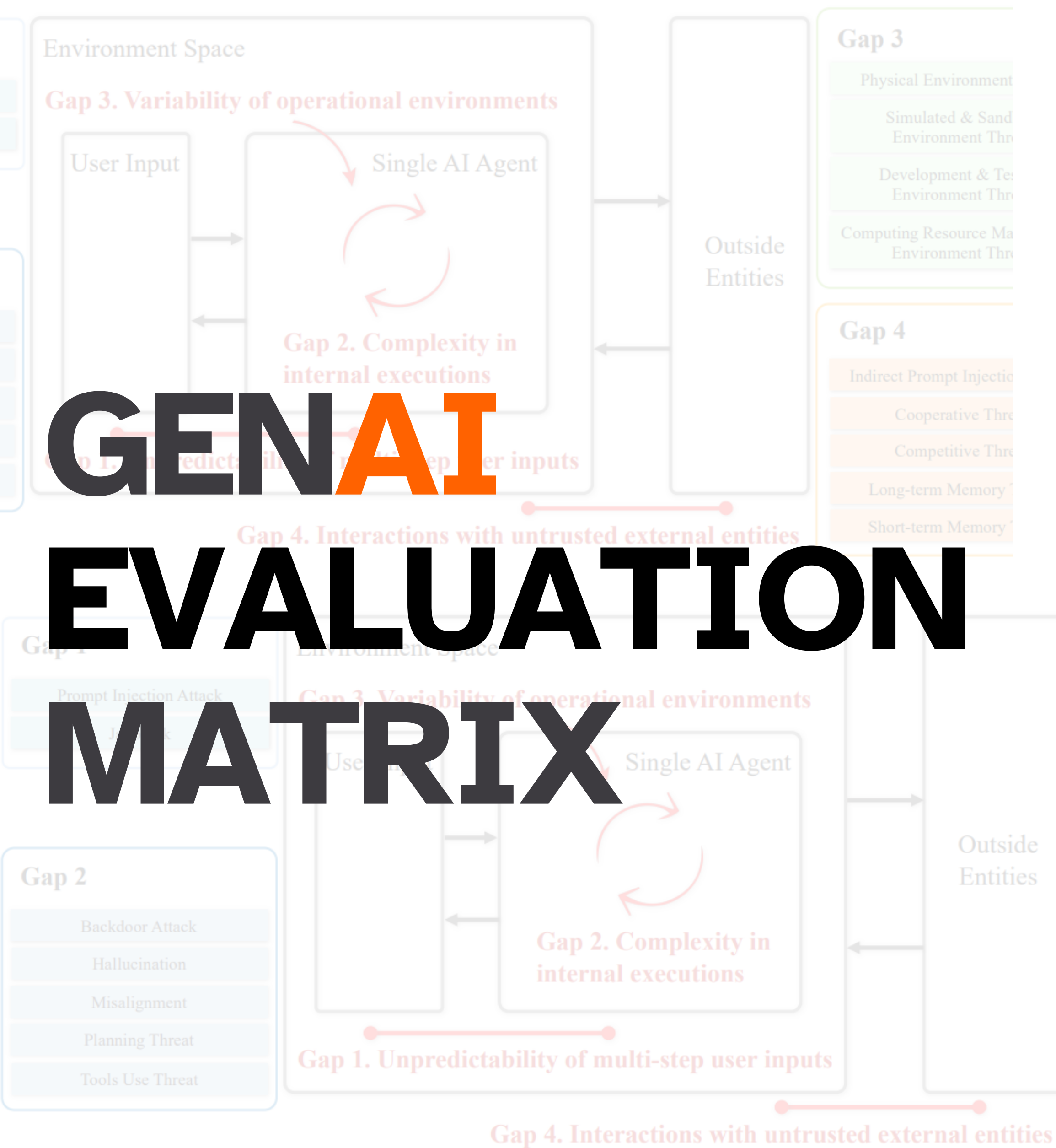


GENAI EVALUATION MATRIX



Introduction

In traditional ML, we check if the model's predictions are right or wrong using a confusion matrix.

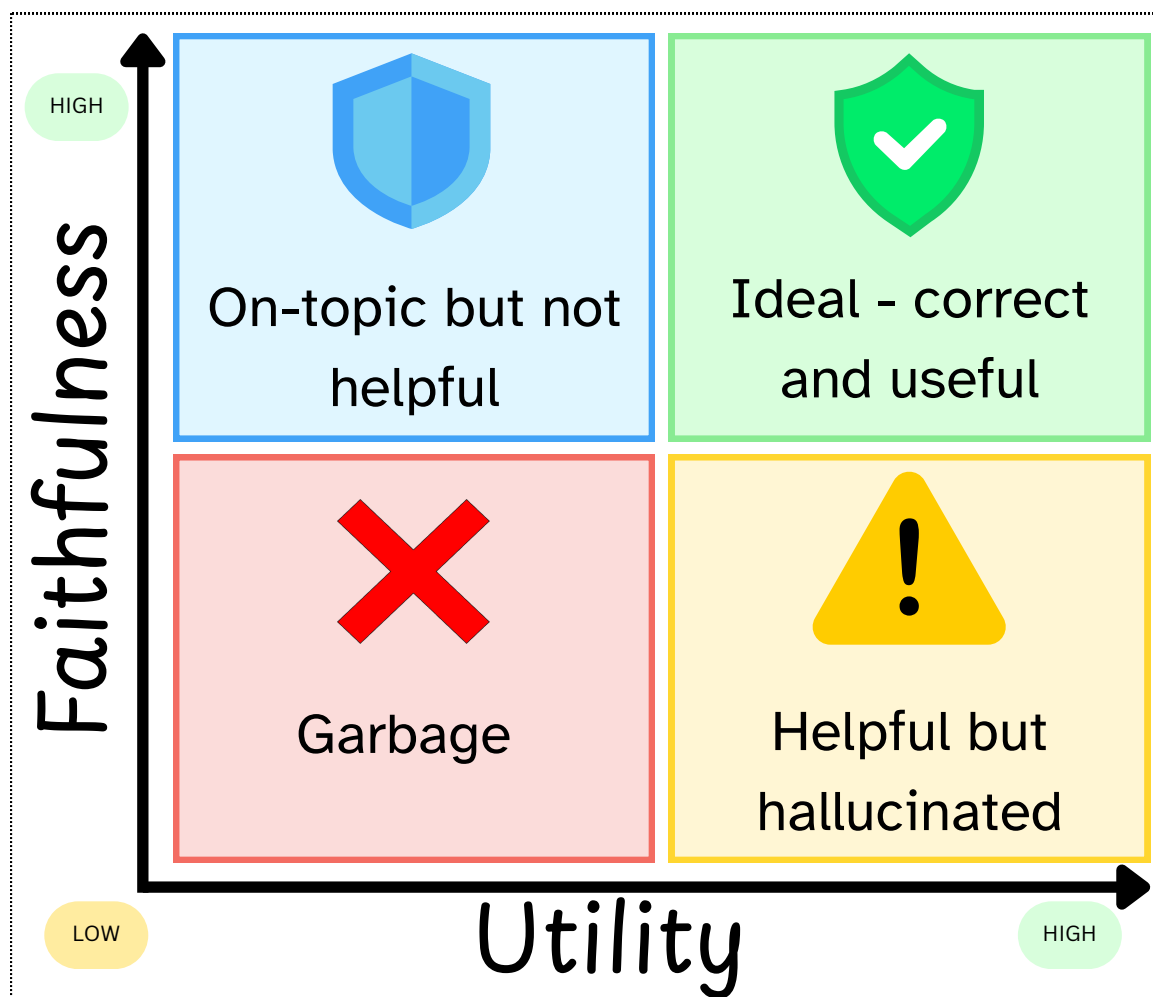
		Ground truth / label <i>Gold standard / Reference test</i>		
		Condition Positive	Condition Negative	
Index test	Predicted Positive	True Positive <i>TP</i>	False Positive <i>FP</i>	Precision <i>Positive predictive value</i> $\frac{TP}{(TP + FP)}$
	Predicted Negative	False Negative <i>FN</i>	True Negative <i>TN</i>	Negative predictive value $\frac{FN}{(FN + TN)}$
		Recall Sensitivity $\frac{TP}{(TP + FN)}$	Specificity $\frac{TN}{(TN + FP)}$	Accuracy $\frac{TP + TN}{(TP + FP + TN + FN)}$
		Source		

But with GenAI, a model might sound confident, write beautifully, and still make things up .

So what if we borrowed this idea of a **confusion matrix** for classification problems and reimaged it for GenAI?

Think of the GenAI Evaluation Matrix as a framework to help you see where your model shines and where it hallucinates.

Faithfulness vs. Utility



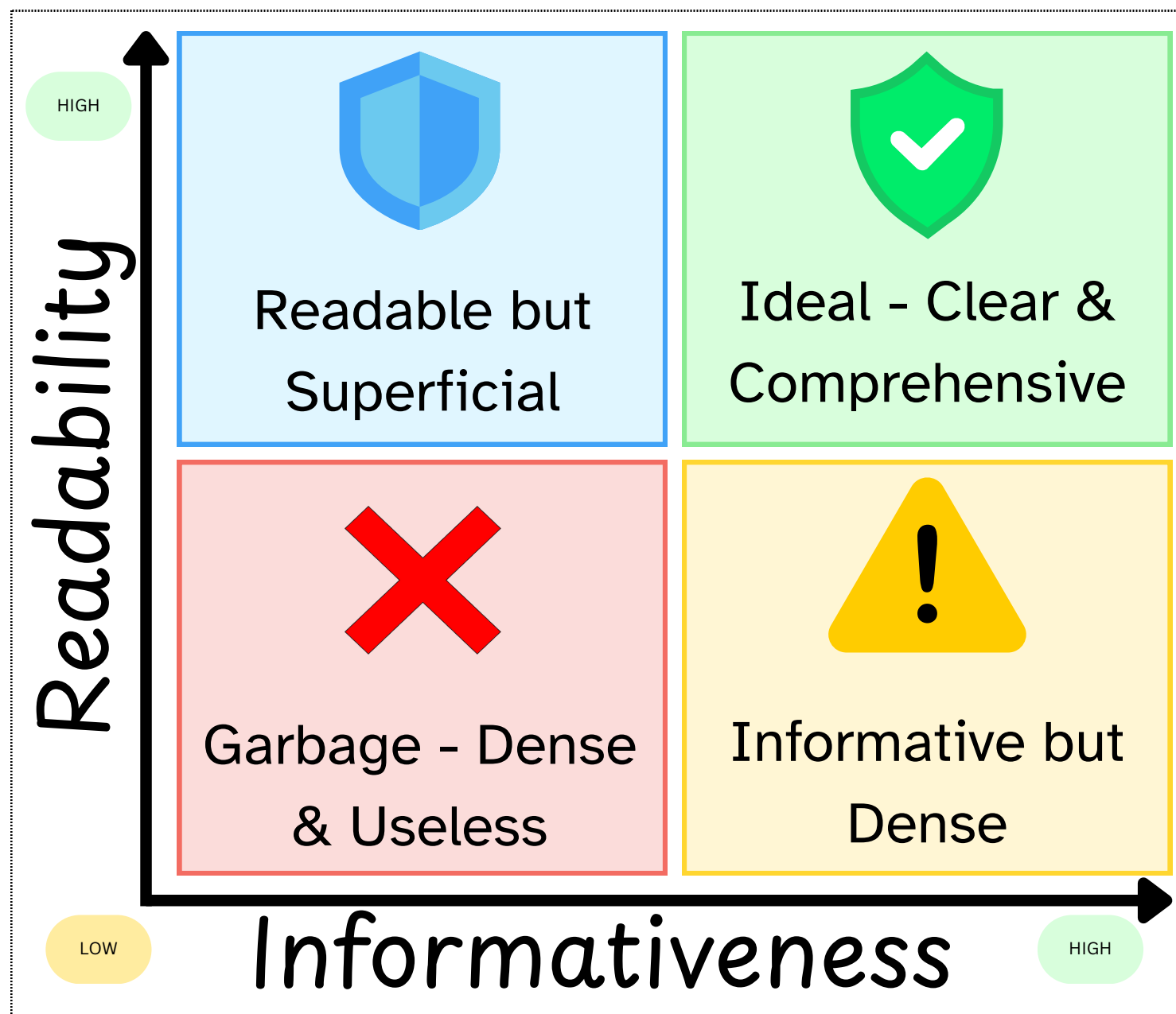
Here's the analogy to confusion matrix for classification model

- True Positive (TP): the idea is clear and informative.
- False Positive (FP): the idea is verbose or confusing — too wordy, adds clutter.
- True Negative (TN): the model keeps it concise where no detail is needed.
- False Negative (FN): the model misses important information that should've been mentioned.

We can compute the derived metrics like the following.

- **Faithfulness Precision:** $TP / (TP + FP)$ → When it says something, how often is it right?
- **Coverage Recall:** $TP / (TP + FN)$ → Of what it should say, how much did it cover?
- **Hallucination Rate:** $FP / (TP + FP)$

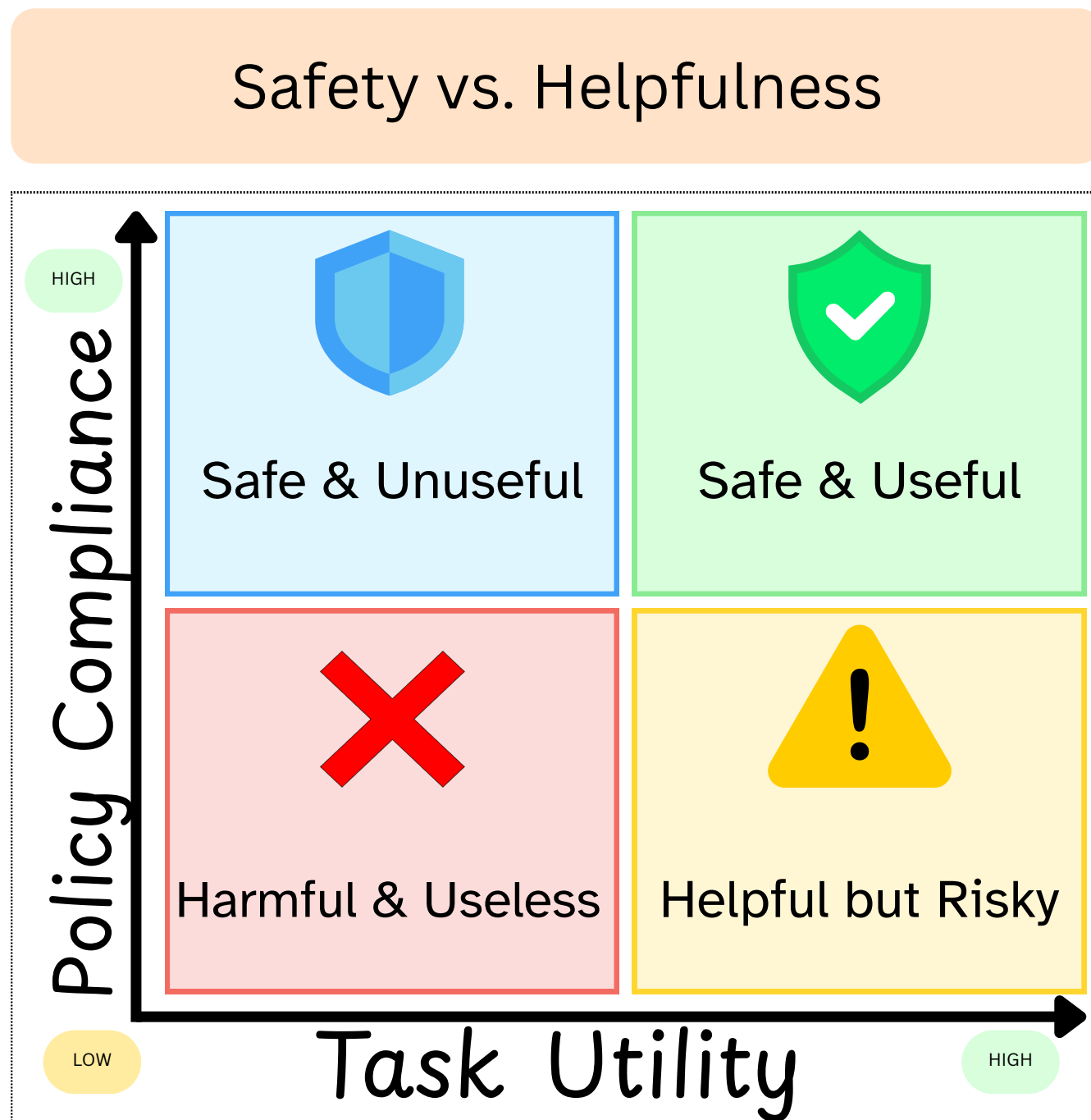
Readability vs Informativeness



To make it more measurable, you can imagine some simple metrics like:

- **Readability Score:** How easy is it to follow?
- **Information Density:** Ratio of unique facts or insights per sentence.
- **Clarity Precision:** Proportion of sentences that are both correct and understandable.
- **Comprehensiveness Recall:** How much relevant information was actually covered.
- **Verbosity Rate:** How often the model says more than needed.

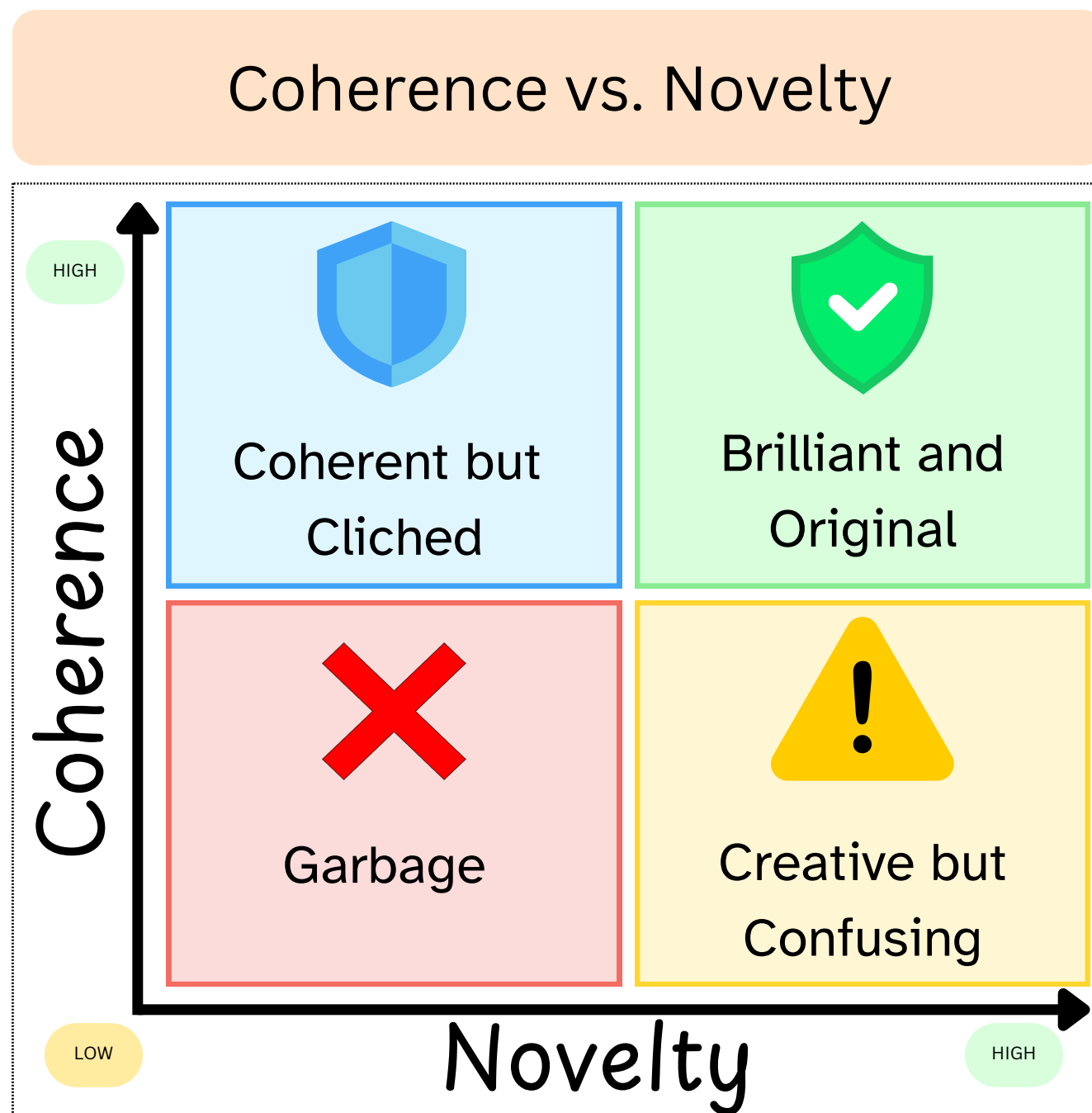
Other Evaluation Metrics



From this, we can define simple metrics:

- **Safety Precision** = $TP / (TP + FP)$ → When it answers, how often is it safe?
- **Helpfulness Recall** = $TP / (TP + FN)$ → Of the safe opportunities to help, how many did it fulfill?
- **Risk Rate** = $FP / (TP + FP)$ → How often does it cross safety boundaries while trying to help?

Other Evaluation Metrics



We can define these metrics:

- **Citation Precision** = $TP / (TP + FP)$ → When it cites, how often is the citation actually valid?
- **Faithfulness Recall** = $TP / (TP + FN)$ → Of all verifiable claims, how many are properly cited?
- **Hallucinated Citation Rate** = $FP / (TP + FP)$ → How often does it cite something wrong or unrelated?

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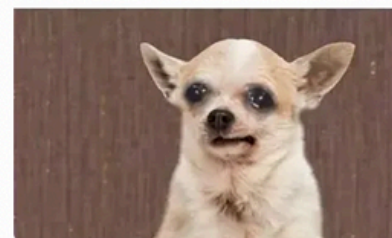
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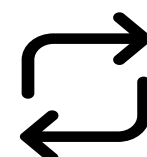
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